Kanwar Pannu

1 Introduction

This is an informal report for the acoustic signal classification project. The goal of this project is to classify acoustic signals using machine learning techniques into "Drummy" or "Tight". The dataset used for this project is divided into 2 phases: Phase 1 and Phase 2.

2 Stage 1

2.1 Data Analysis

2.1.1 Phase 1

Phase 1 consists of **3,309 recordings** (**2,212 drummy** sounds and **1,097 tight sounds**). Recordings in this phase have a duration of **0.75 seconds** and **36000 samples per recording**. The recordings are sampled at a rate of **48 kHz**. The dataset is not balanced, with number of drummy sounds twice that of tight sounds.

Findings

- Did a quick PCA analysis (first standardizing the data) and plotted the first 2 principal components. The 3D scatter plot (PC1 vs PC2 vs Label) shows that drummy and tight sounds are not linearly separable, and that tight sounds form a cluster in the middle of drummy sounds, while drummy sounds are more spread out.

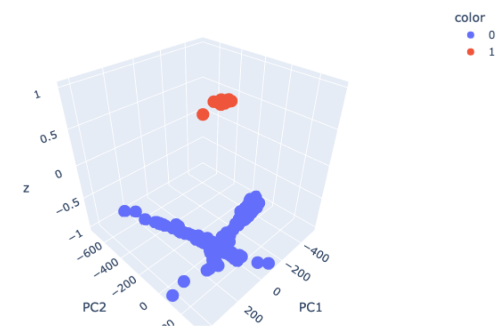
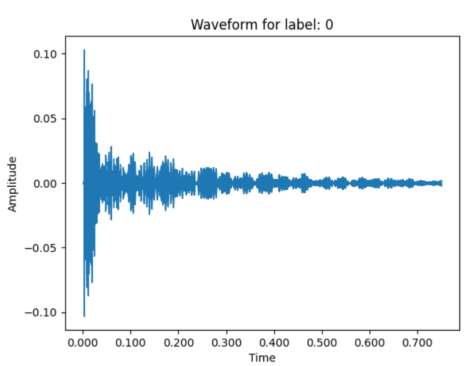
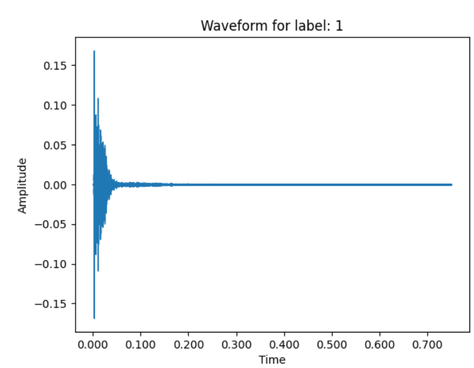


Figure 1: 2 Component PCA Analysis of Phase 1 Data

- Waveform visualization shows that drummy sounds dampen relatively slowly and have a longer decay time, while tight sounds dampen quickly and have a shorter decay time. This could suggest that sounds oscillated more in the hollow structure of the roof that produce the drummy sounds.



((a)) Drummy Waveform Visualization (Phase 1)



((b)) Tight Waveform Visualization (Phase 1)

Figure 2: Waveform visualizations for different drum styles in Phase 1

- Explained variance ratio vs Number of components gives the following results:
  Number of components retaining 95% variance: 396
  Number of components retaining 80% variance: 87
  Number of components retaining 70% variance: 48

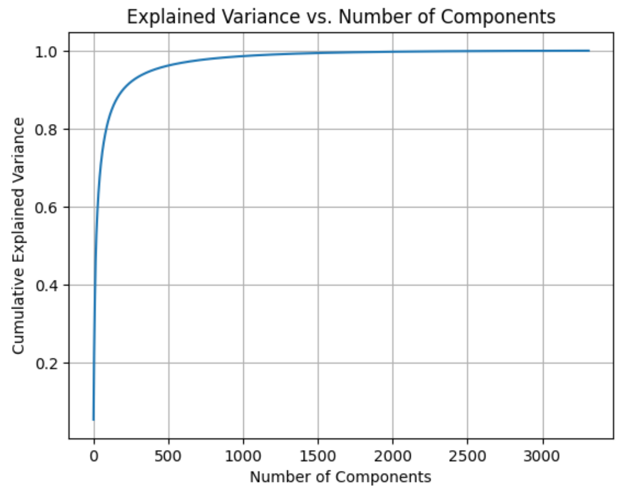


Figure 3: Explained Variance Ratio vs Number of Components (Phase 1)

2.1.2 Phase 2

Phase 2 consists of **9,923 recordings** (from which **1,483** recordings were **corrupted with NaN value**s, making it 8,440 recordings; further, there are **some mis-hits** as well that need to be cleaned later for feature extraction to be done on them). Finally, there are **3,856 drummy sounds and 3,124 tight sounds (including the mis-hits)**. Each recording has 4800 samples and a duration of **0.1 seconds**. The recordings are sampled at a rate of **48 kHz**. The dataset is more balanced than Phase 1, with number of drummy sounds slightly higher than tight sounds.

Findings

- Performed a quick PCA analysis (after standardizing the data) and plotted the first two principal components. The 3D scatter plot (PC1 vs PC2 vs Label) indicates that drummy and tight sounds are not linearly separable; both classes are intermixed in the feature space.

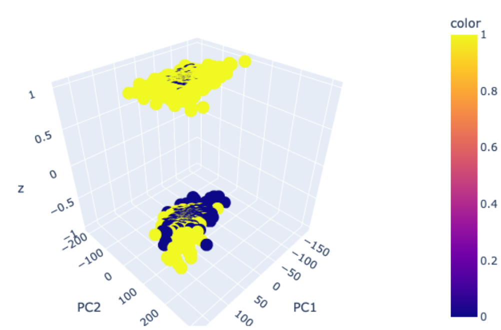
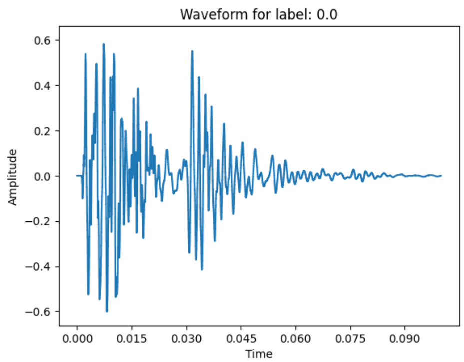
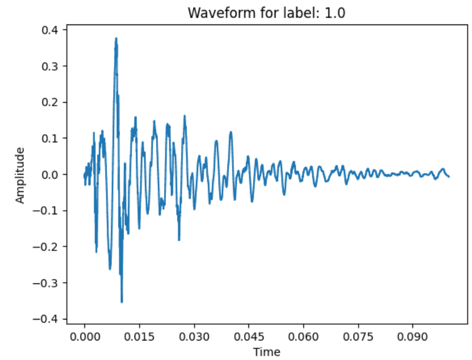


Figure 4: 2 Component PCA Analysis of Phase 2 Data

- Waveform visualizations show that drummy sounds decay more slowly and have a longer decay time, while tight sounds decay quickly and have a shorter decay time. This suggests that drummy sounds may result from greater oscillation in the hollow structure of the roof.



((a)) Drummy Waveform Visualization (Phase 2)



((b)) Tight Waveform Visualization (Phase 2)

Figure 5: Waveform visualizations for different drum styles in Phase 2

- The explained variance ratio versus number of components yields the following results:
  Number of components retaining 95% variance: 242
  Number of components retaining 80% variance: 99
  Number of components retaining 70% variance: 65

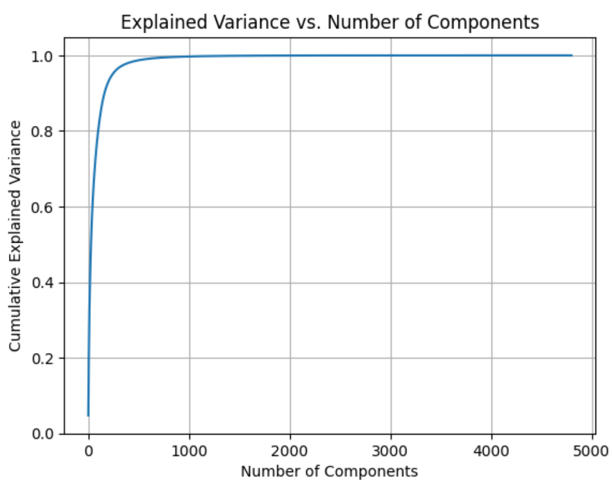


Figure 6: Explained Variance Ratio vs Number of Components (Phase 2)

3 Feature Extraction

3.1 PCA

3.1.1 Phase 1 Data

PCA / **MLP** Used PCA to reduce dimensionality and then classified using MLP with one hidden layer of varying number of neurons. The best results were obtained with 1000 PCA components and 64 hiddens neurons with test accuracy of 0.838188. Below is the compariosn chart.

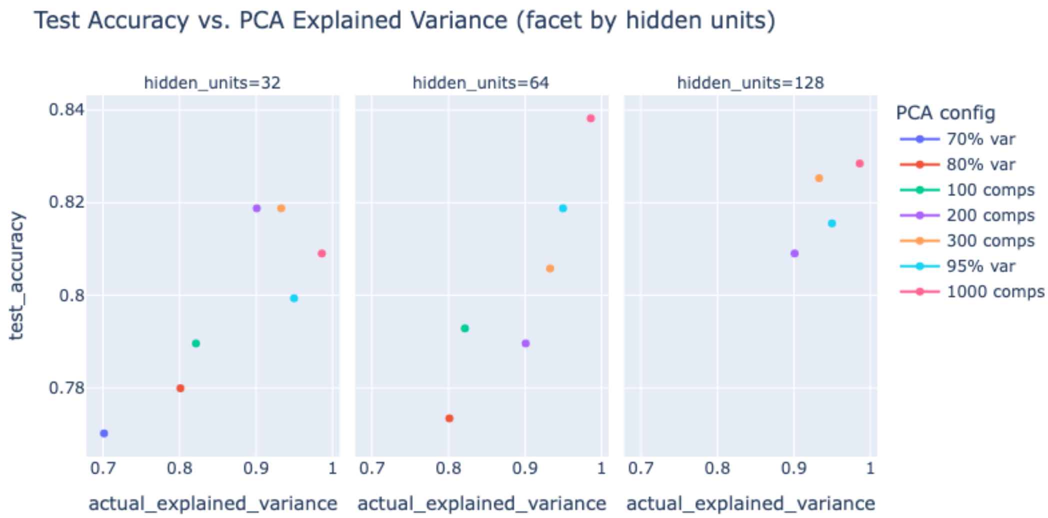
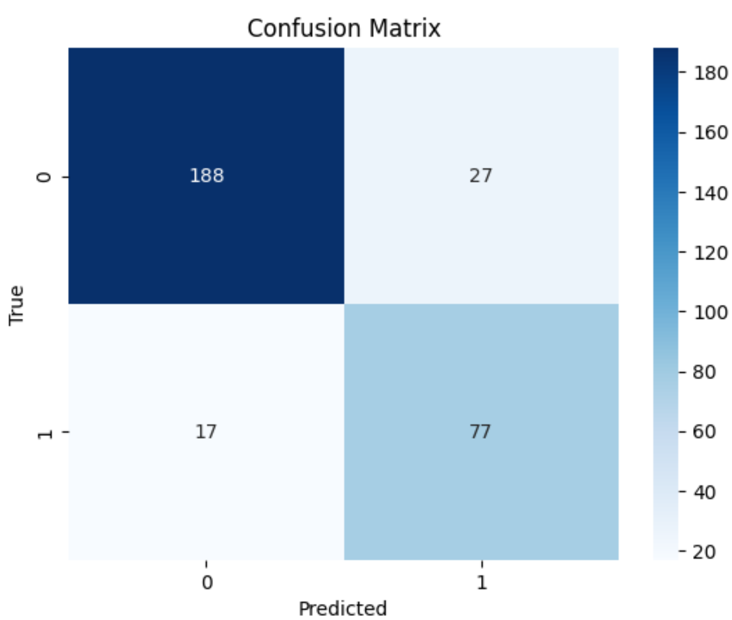


Figure 7: PCA + MLP Results (Phase 1)

PCA / SVM Used PCA to reduce dimensionality and then classified using SVM with **RBF kernel**. The best results were obtained with 1000 PCA components and a C value of 100 with test accuracy of averaging around 0.86 if we do multiple runs. Below is confusion matrix for the best run and score chart.



((a)) PCA + SVM Confusion Matrix (Phase 1)

	precision	recall	f1-score	support
0	0.92	0.87	0.90	215
1	0.74	0.82	0.78	94
accuracy			0.86	309
macro avg	0.83	0.85	0.84	309
weighted avg	0.86	0.86	0.86	309

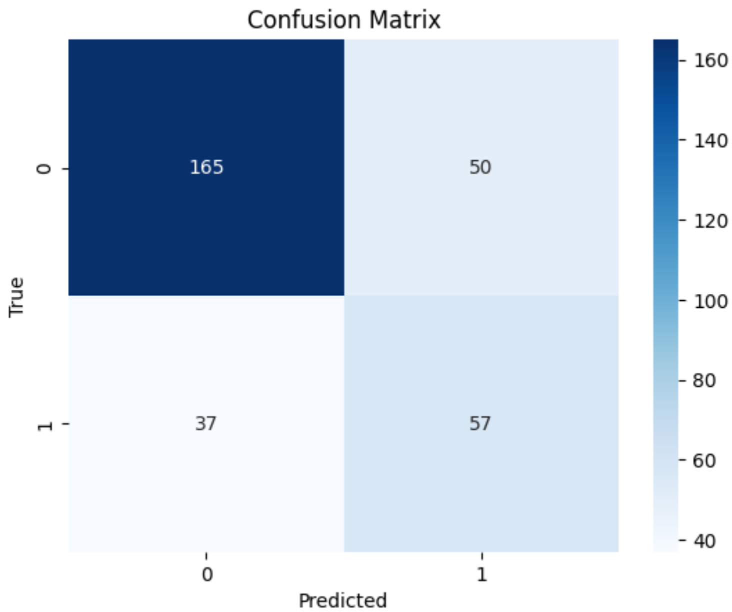
((b)) PCA + SVM Score Chart (Phase 1)

Figure 8: PCA + SVM evaluation results for Phase 1: (a) confusion matrix, (b) score chart.

3.2 LDA

3.2.1 Phase 1 Data

LDA for multiclass classification Used LDA to reduce dimensionality and classification. Below is the confusion matrix and score chart for the best run with 1 components.



((a)) LDA Confusion Matrix (Phase 1)

	precision	recall	f1-score	support
0	0.82	0.77	0.79	215
1	0.53	0.61	0.57	94
accuracy			0.72	309
macro avg	0.67	0.69	0.68	309
weighted avg	0.73	0.72	0.72	309

Accuracy: 0.7184466019417476

((b)) LDA Score Chart (Phase 1)

Figure 9: LDA evaluation results for Phase 1: (a) confusion matrix, (b) score chart.

3.3 Time-Domain Features

These features are computed on short frames of the raw waveform and capture simple temporal properties.

3.3.1 Zero Crossing Rate (ZCR)

Extracted ZCR using 128-sample frames and 32-sample hops. Classification results:

- **SVM (RBF, C=100)**: Test accuracy = 0.7799
- **MLP (1-layer)**: Test accuracy = 0.7832
- **MLP (2-layer)**: Test accuracy = 0.8058
- **MLP (3-layer)**: Test accuracy = 0.8155

3.3.2 RMS Energy

Extracted frame-wise RMS energy with 36-sample frames and 36-sample hops. Classification results:

- **SVM (RBF, C=100)**: Test accuracy = 0.8285
- **MLP (1-layer)**: Test accuracy = 0.7832
- **MLP (2-layer)**: Test accuracy = 0.8123
- **MLP (3-layer)**: Test accuracy = 0.8414

3.3.3 Amplitude Envelope

Computed the amplitude envelope on 128-sample frames with 16-sample hops. Classification results:

- **SVM (RBF, C=100)**: Test accuracy = 0.7411
- **MLP (1-layer)**: Test accuracy = 0.7638
- **MLP (2-layer)**: Test accuracy = 0.7961
- **MLP (3-layer)**: Test accuracy = 0.8123

3.4 Combined Time-Domain Features

Concatenated multiple time-domain features frame-wise and retrained the MLP.

- **Envelope + RMS**: Test accuracy = 0.8317
- **Envelope + ZCR**: Test accuracy = 0.8317
- **RMS + ZCR**: Test accuracy = 0.8447
- **Envelope + RMS + ZCR**: Test accuracy = 0.8641

3.5 Leave-One-Mine-Out Evaluation

I performed a leave-one-mine-out cross-validation using the combined feature vector (Envelope + RMS + ZCR) and an MLP with 3 layers (1024 hidden units, dropout 0.5). Figure **Leave One Out** shows test accuracy for each held-out mine.

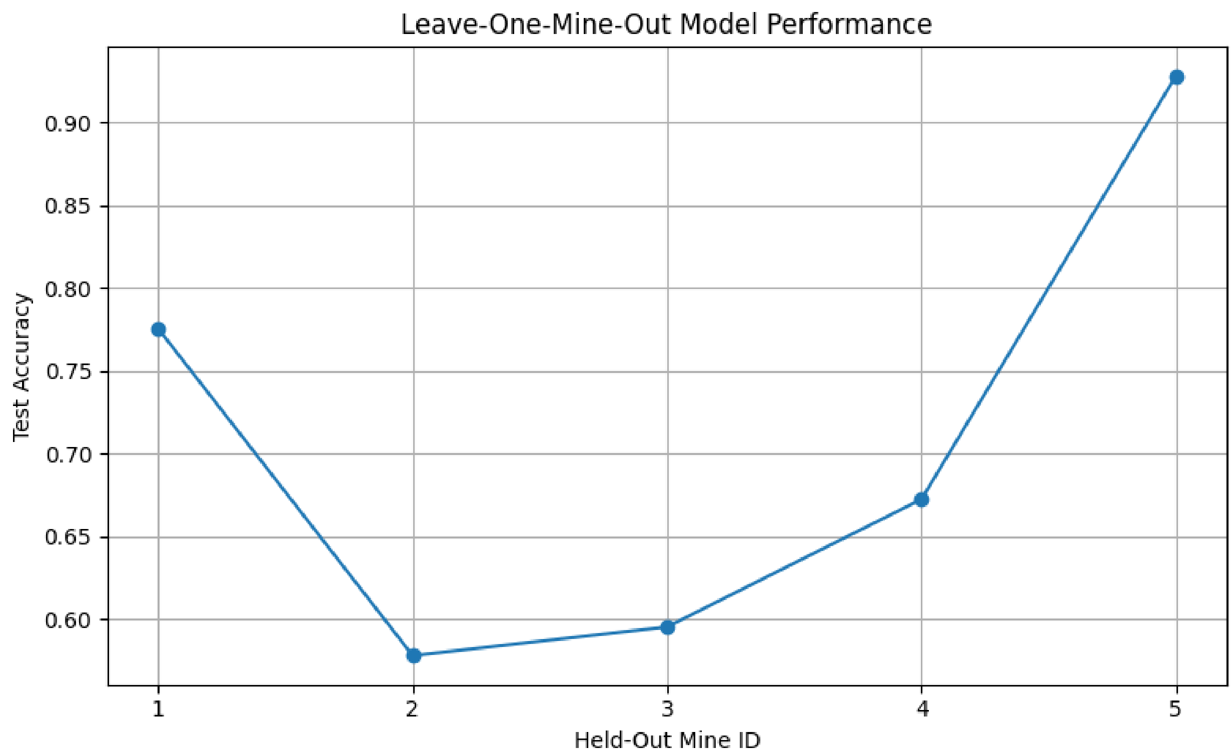


Figure 10: Leave-One-Mine-Out Performance

Average held-out accuracy: **0.7754** (drummy vs tight across mines).

3.6 Continuous Wavelet Transform (CWT)

I applied a CWT with the Complex Morlet wavelet (scales=64, 400 frames) and trained a WaveNet-style CNN.

- WaveNet classifier test accuracy = 0.6958

4 Discussion

The RMS energy and combined time-domain features outperform single features. CWT features yield lower accuracy under the current architecture, suggesting further tuning or deeper models are needed.